Jiayi (Chesley) Guo

Berkeley, CA

217-328-6257 chesleyguo@berkeley.edu linkedin.com/in/jg1023 chesleyguo.github.io

Education

University of California, Berkeley **2025 – Dec 2026 (Expected)**

Master of Development Engineering (MDE), AI/Data Analytics Track; GPA: 3.82 *Berkeley, CA*

- **Core Courses:** Experiments & Causal Inference; Research Design & Applications for Data and Analysis; Big Data & Development; Digital Transformation of Development; Design, Evaluate & Scale Development Technologies; Critical Systems of Development; Front-End Web Architecture; Public Interest Technology Clinic
- **Capstone:** Selection-Aware Simulation-Based Inference for Landfill Methane from Sparse Airborne Surveys
- **Honors:** NSF NRT Fellow in Digital Transformation of Development (DToD), 2025

University of Illinois Urbana-Champaign **2023 – 2024**

Bachelor of Science (BS) in Information Sciences, Data Science Track; GPA: 3.91 *Urbana, IL*

- **Core Courses:** Data Visualization; Research Design in Information Sciences; Programming for Information Problems; Analytical Foundations of Information Problems; Reading and Writing Data; Introduction to HCI; Ethics & Policy for Data Science; Data Structures (transfer credit); Advanced Software Design (transfer credit)
- **Honors:** Edmund J. James Scholar (2023–2024)
  Dean's List (Fall 2023)
  Phi Kappa Phi Honor Society (2024)
  WeCU Scholar, 200+ hours of volunteer service (2024)
  Top 3 Finalist, Equitable Access to Healthcare Challenge, MIT Policy Hackathon (2023)
  Competition Expert, Kaggle (2023)

Xi'an Jiaotong-Liverpool University **2020 – 2022**

Undergraduate study in Bioinformatics (two years); transferred to UIUC *Suzhou, China*

- **Core Courses:** Cellular & Molecular Biology; Introduction to Biochemistry; Introduction to Evolution & Genetics; Algorithmic Foundations & Problem Solving; Introduction to Databases; Introduction to Programming in Java; Experimental Design & Analysis; Introduction to Statistics; Linear Algebra; Multivariable Calculus
- **Honors:** Meritorious Winner, Interdisciplinary Contest in Modeling (ICM, 2022)
  National Second Prize, China Undergraduate Mathematical Contest in Modeling (2021)
  District Third Prize, Asia-Pacific Mathematical Contest in Modeling (2020)

Publications

- **Guo, J.** Retracted but not forgotten: A matched-control audit of how consumer large language models handle retracted clinical trials. *Submitted to Scientometrics.*
- **Guo, J.**, & Hughes, A. The effect of automated speed enforcement on driving behavior: Evidence from Oakland's speed safety camera pilot. *Working paper.*
- **Guo, J.**, & Agogino, A. M. Counted twice: Selection-aware inference of facility-level landfill methane emissions from sparse airborne surveys. *Manuscript in preparation for Environmental Science & Technology.*
- Lu, H., Gu, L., **Guo, J.**, Xu, L., et al. (2023). Machine learning classification and recovery prediction of COVID-19 cough sounds. *SSRN (Preprints with The Lancet).* SSRN 4439141
- Shao, Y., **Guo, J.**, Li, Z., Yan, W., & Zhu, Z. (2022). Optimization of seaport companies' D&A systems based on system dynamics and evaluation models. *Proceedings of SPIE*, 12340, 1234003. doi:10.1117/12.2652356

Additional Training

MITx MicroMasters Program in Statistics and Data Science **2024**

Completed all four courses with verified certificates: Probability; Statistics; ML with Python; Data Analysis *Online*

Oxford Machine Learning Summer School (OxML) **2023**

ML × Health Track *Oxford, UK*

SJTU–THU–PKU Computer Science International Summer School **2023**

Courses: Artificial Intelligence; Cryptography in Blockchain *Shanghai, China*

Healthcare Big Data Summer Program, Duke Kunshan University **2022**
Courses: Health Data Analytics; ML in Health Sciences; Causal Inference & RCTs; Real-World Evidence *Kunshan, China*

Data Analysis for Life Sciences Professional Certificate, Harvard University **2022**
Courses: Statistics and R; Linear Models; Statistical Inference; High-Dimensional Data Analysis *Online*

Independent Research

Retracted but Not Forgotten: Consumer LLMs and Retracted Clinical Trials **Aug 2026 – Present**
Sole Author; manuscript submitted to Scientometrics *Berkeley, CA*

- Designed a matched-control audit of ChatGPT, Gemini, and Claude on 100 retracted clinical trials from Retraction Watch, each paired with a control matched on topic, year, and design; collected 3,600 responses across 3 question framings and 3 conditions.
- Validated an LLM coder against hand coding ($\kappa = 0.96$) and modeled outcomes with GEE.
- Found that about two-thirds of answers presented retracted findings as valid, no different from controls; a retraction notice cut errors to $\leq$2%, while web search had mixed effects (ChatGPT 66% $\rightarrow$ 35%, Gemini 66% $\rightarrow$ 82%). Code and data: github.com/chesleyguo/llm-retracted-clinical-evidence

Research Experience

Selection-Aware Inference for Landfill Methane **Jul 2026 – Present**
Research Assistant, BEST Lab, UC Berkeley (Advisor: Prof. Alice Agogino) *Berkeley, CA*

- Developing selection-aware simulation-based inference for facility-level landfill methane from sparse airborne surveys (AVIRIS-NG, GAO; 107 U.S. landfills, 593 flight days): a likelihood conditioned on at least one detection with instrument detection limits, cross-fitted hierarchical priors, and a neural posterior estimator (PyTorch) checked against exact MCMC.
- In simulation, averaging only detected overpasses overstated emissions at single-overpass sites by 41%; the neural estimator reduced the bias to -3%, with 90% intervals covering the truth 90–91% of the time.
- Froze the analysis configuration before real-data results and kept a public decision log. Code: github.com/chesleyguo/landfill_methane_dev
- Audited EPA's Landfill Methane Outreach Program register (2,641 landfills; public explorer) and built a methane dashboard now used by Squishy Robotics to contact landfill operators (20 to date).

Causal Impact of Automated Speed Enforcement **Aug 2026 – Present**
Student Team Data Scientist, Berkeley I School × Oakland DOT (Prof. Alex Hughes) *Berkeley, CA*

- Estimating the effect of Oakland's 18-camera speed safety pilot with a regression discontinuity in time at two rollout cutoffs, separating the effect of camera warnings from the added effect of fines.
- Analyzing speed counts from 2024–2026 and camera violation records since the January 2026 launch in a secure GCP environment under CPRA/CCPA compliance.

OpenTrust Lab, Shanghai Artificial Intelligence Laboratory **Jun 2024 – Aug 2024**
Research Intern; Supervisor: Dr. Jing Shao, Research Scientist *Shanghai, China*

- Assisted with safety evaluations of large language models on the lab's benchmarks (SALAD-Bench, MM-SafetyBench).
- Supported organization of the WAIC 2024 Security Forum and the TiFA Workshop at ICML 2024.

School of Information Sciences, University of Illinois Urbana-Champaign **Jan 2024 – May 2024**
Research Assistant; Supervisor: Prof. Yun Huang *Urbana, IL*

- Conducted literature review and comparative analysis on cross-cultural Human-Computer Interaction (HCI) designs for academic information-seeking behaviors.
- Assisted in the conceptual design and requirement specification for an AI-powered chatbot tailored to enhance accessibility and efficiency in academic search and literature discovery for a global user base.
- Proposed and outlined key features for the chatbot, including multilingual support, context-aware query interpretation, and integration with digital library APIs, to address diverse user needs in scholarly environments.

Vanke School of Public Health, Tsinghua University **Nov 2022 – Mar 2023**
Research Assistant; Supervisor: Prof. Kun Tang *Beijing, China (Remote)*

- Collected, cleaned, and processed textual data on medical cooperation memorandums and IHME disease burden metrics (DALYs) for Belt and Road countries since 2016 (Africa, South America, Southeast Asia) to support the team's downstream analysis.

Global Health Institute, Duke Kunshan University **Mar 2022 – Aug 2022**

Health Research Data Analyst Intern; Supervisor: Prof. Chenkai Wu *Kunshan, China*

- Conducted a comprehensive analysis using descriptive statistics, principal component analysis, and multinomial regression to examine the relationship between 249 metabolites from the UK Biobank and the impact of sarcopenia on aging; built mortality risk predictions using the Cox proportional hazards model.

Shanghai Key Laboratory of Multidimensional Information Processing **Oct 2021 – Dec 2022**

Research Assistant; Supervisor: Prof. Jiangang Chen, East China Normal University *Shanghai, China*

- Collected cough acoustics data from 78 subjects at a tertiary hospital in Shanghai; performed data processing and cleaning to support the identification of COVID-19 patients.
- Leveraged voice activity detection, noise reduction, and data augmentation techniques to improve data quality, expanding the dataset to 535 audio samples; extracted diverse audio features and their statistical characteristics using `librosa` and `scipy`.
- Constructed prognostic models integrating multiple machine learning algorithms (e.g., Random Forest, XGBoost, SVM) to balance detection rate and prediction of time to recovery; reduced prognostic time error to within 3 days and achieved 93.3% accuracy with an ensemble XGBoost and Random Forest model.

Work Experience

Tencent **May 2025 – Aug 2025**

Product Manager Intern *Shanghai, China*

- Analyzed product usage data for Tencent's Real-Time Communication (RTC) suite (e.g., Conversational AI, Call, Conference, Live Streaming) to identify and prioritize feature improvements for a global user base.
- Ran A/B tests across marketing campaigns to compare the effectiveness of content strategies.

ByteDance (TikTok) **Nov 2024 – Jan 2025**

AI Strategy Product Manager Intern *Shanghai, China*

- Worked with algorithm engineers to improve video understanding and text generation LLMs, focusing on description accuracy and hallucination reduction.
- Drafted product requirement documents (PRDs) and defined core functionalities for LLMs, while designing and refining evaluation metrics such as accuracy, semantic relevance, and logical reasoning.

CSC Financial Co., Ltd. **Jul 2023 – Aug 2023**

Financial Planning and Analysis Analyst Intern *Changzhou, China*

- Built financial models and forecasts to support budget creation and the firm's strategic planning process.
- Conducted competitor and market analyses to identify growth opportunities, and presented findings and recommendations to senior management.

Teaching & Mentoring

- **Research Mentor**, Undergraduate Research Apprentice Program (URAP), UC Berkeley 2026 – Present
  Mentoring 2 undergraduates on their independent research projects in AI/ML for methane monitoring.
- **Senior Mentor**, Xi'an Jiaotong-Liverpool University Affiliated School 2022 – 2023
  Taught STEM enrichment courses to a class of ∼50 high school students.
- **Student Representative**, School of Science, Xi'an Jiaotong-Liverpool University 2021 – 2022
  Mentored junior undergraduates on coursework and academic planning.